Assignment Number: 1 (a & b)

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Assignment Title: Design and implement Parallel Breadth First Search and Depth First Search based on existing algorithms using OpenMP. Use a Tree or an undirected graph for BFS and DFS.

CODE

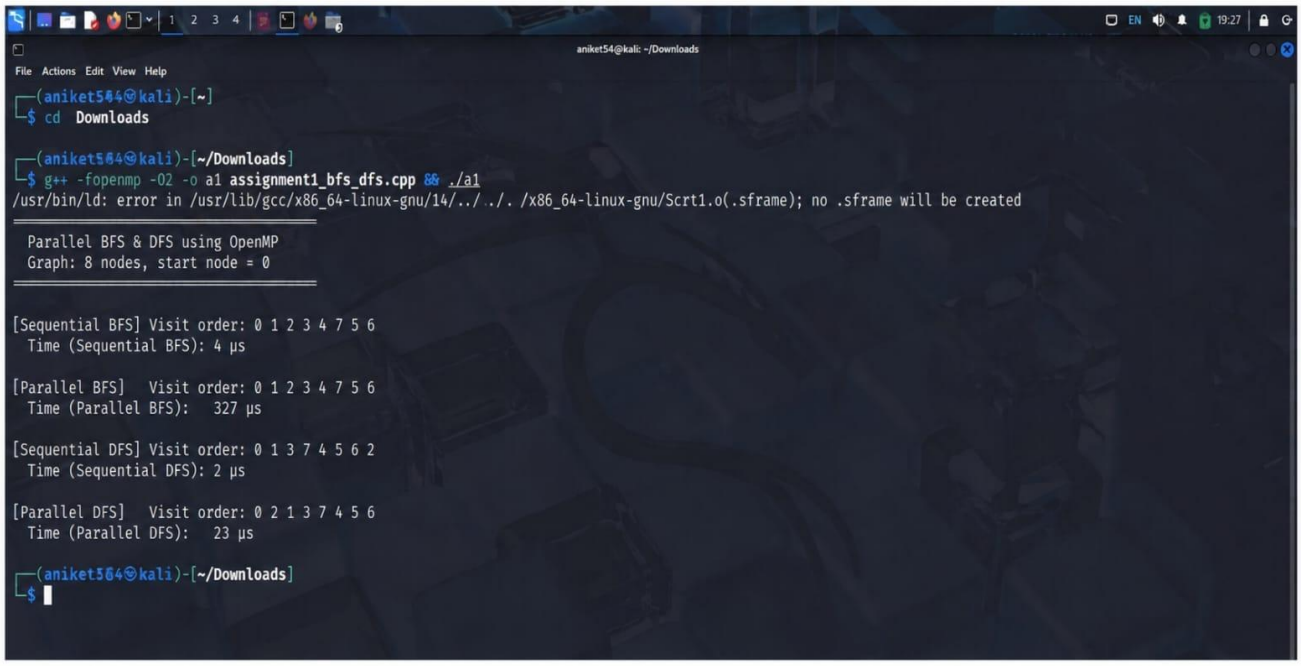
```
// ============================================================
// Undirected Graph representation using Adjacency List
// ============================================================

#include <iostream>
#include <vector>
#include <queue>
#include <stack>
#include <omp.h> #include
<chrono>
using namespace std;

class Graph { public:     int
V;     vector<vector<int>>
adj;

   Graph(int v) : V(v), adj(v) {}

   void addEdge(int u, int v)
{         adj[u].push_back(v);
adj[v].push_back(u); // undirected
   }

   // ——— Sequential BFS ——————————————————————————————————————
void BFS_Sequential(int start) {         vector<bool> visited(V, false);         queue<int> q;
visited[start] = true;         q.push(start);

       cout << "\n[Sequential BFS] Visit order: ";
while (!q.empty()) {             int node = q.front();
q.pop();             cout << node << " ";             for
(int nbr : adj[node]) {                 if (!visited[nbr])
{                     visited[nbr] = true;
q.push(nbr);
               }
           }
       }
       cout << endl;
   }
  // ——— Parallel BFS (level-synchronised) ——————————————————————
void BFS_Parallel(int start) {         vector<bool> visited(V, false);
vector<int> current_level, next_level;
```

```
    visited[start] = true;
    current_level.push_back(start);

     cout << "\n[Parallel BFS]   Visit order: ";
while (!current_level.empty()) {            //
Print current level nodes
        for (int node : current_level) cout << node << " ";

        next_level.clear();

        // Process all nodes in current level in parallel
        #pragma omp parallel
        {
          vector<int> local_next;
          #pragma omp for nowait schedule(dynamic)
for (int i = 0; i < (int)current_level.size(); i++)
{                 int node = current_level[i];                  for
(int nbr : adj[node]) {                    bool already;
                  #pragma omp critical
                  {
                    already = visited[nbr];
                    if (!already) visited[nbr] = true;
                  }
                  if (!already) local_next.push_back(nbr);
              }
          }
          #pragma omp critical
          next_level.insert(next_level.end(), local_next.begin(), local_next.end());
        }
        current_level = next_level;
    }
    cout << endl;
  }

  // ——— Sequential DFS ——————————————————————————————————————————
void DFS_Sequential(int start) {       vector<bool> visited(V, false);
    stack<int> st;
    st.push(start);

    cout << "\n[Sequential DFS] Visit order: ";
```

```
        while (!st.empty())
{           int node = st.top();
st.pop();           if (!visited[node])
{               visited[node] = true;
            cout << node << " ";            for (int i =
(int)adj[node].size() - 1; i >= 0; i--)                 if
(!visited[adj[node][i]])
st.push(adj[node][i]);
          }
       }
       cout << endl;
    }
    // ——— Parallel DFS (parallel neighbour exploration) ——————————
void DFS_Parallel_Helper(int node, vector<bool>& visited) {
        #pragma omp critical
{ visited[node] = true; }
        cout << node << " ";

        #pragma omp parallel for schedule(dynamic)
for (int i = 0; i < (int)adj[node].size(); i++)
{           int nbr = adj[node][i];           bool v;
          #pragma omp critical
{ v = visited[nbr]; }
          if (!v) DFS_Parallel_Helper(nbr, visited);
        }
    }

    void DFS_Parallel(int start)
{       vector<bool> visited(V, false);
        cout << "\n[Parallel DFS]   Visit order: ";
DFS_Parallel_Helper(start, visited);
        cout << endl;
    }
};

int main() {  // Build a sample undirected graph with 8 nodes
    //  0 - 1 - 3 - 7
    //  |   |
    //  2   4 - 5
    //      |
    //      6
```

```
    Graph g(8);
    g.addEdge(0, 1); g.addEdge(0, 2);
    g.addEdge(1, 3); g.addEdge(1, 4);
    g.addEdge(3, 7); g.addEdge(4, 5);
    g.addEdge(4, 6);

    cout << "======================================" << endl;
    cout << "  Parallel BFS & DFS using OpenMP    " << endl;
    cout << "  Graph: 8 nodes, start node = 0     " << endl;
    cout << "======================================" << endl;

    // ── BFS timing ──────────────────────────────────────────────────────────
auto t1 = chrono::high_resolution_clock::now();          g.BFS_Sequential(0);          auto t2 =
chrono::high_resolution_clock::now();    cout << "  Time (Sequential BFS): "
        << chrono::duration_cast<chrono::microseconds>(t2-t1).count() << " µs\n";

   t1 = chrono::high_resolution_clock::now();
g.BFS_Parallel(0);
    t2 = chrono::high_resolution_clock::now();
cout << "  Time (Parallel BFS):   "
         << chrono::duration_cast<chrono::microseconds>(t2-t1).count() << " µs\n";

    // ── DFS timing ──────────────────────────────────────────────────────────
t1 = chrono::high_resolution_clock::now();     g.DFS_Sequential(0);     t2 =
chrono::high_resolution_clock::now();     cout << "  Time (Sequential DFS): "
         << chrono::duration_cast<chrono::microseconds>(t2-t1).count() << " µs\n";

    t1 = chrono::high_resolution_clock::now();
g.DFS_Parallel(0);     t2 =
chrono::high_resolution_clock::now();
cout << "  Time (Parallel DFS):   "
         << chrono::duration_cast<chrono::microseconds>(t2-t1).count() << " µs\n";

    return 0;
}

// ── Compile & Run ──────────────────────────────────────────────────────────
// g++ -fopenmp -O2 -o a1 assignment1_bfs_dfs.cpp && ./a1
```

OUTPUT


aniket54@kali: ~/Downloads
File Actions Edit View Help
(aniket564@kali)-[~]
$ cd Downloads
(aniket564@kali)-[~/Downloads]
$ g++ -fopenmp -O2 -o a1 assignment1_bfs_dfs.cpp && ./a1
/usr/bin/ld: error in /usr/lib/gcc/x86_64-linux-gnu/14/../../. /x86_64-linux-gnu/Scrt1.o(.sframe); no .sframe will be created
Parallel BFS & DFS using OpenMP
Graph: 8 nodes, start node = 0
[Sequential BFS] Visit order: 0 1 2 3 4 7 5 6
Time (Sequential BFS): 4 µs
[Parallel BFS] Visit order: 0 1 2 3 4 7 5 6
Time (Parallel BFS): 327 µs
[Sequential DFS] Visit order: 0 1 3 7 4 5 6 2
Time (Sequential DFS): 2 µs
[Parallel DFS] Visit order: 0 2 1 3 7 4 5 6
Time (Parallel DFS): 23 µs
(aniket564@kali)-[~/Downloads]
$